UNIVERSITY OF BUEA

TASK 2

FACULTY OF ENGINEERING AND TECHNOLOGY

DEPARTMENT OF COMPUTER ENGINEERING

CEF 440: INTERNET PROGRAMMING AND MOBILE PRORAMMING

TASK1: REQUIREMENTS GATHERING

GROUP 19

Design and Implementation of a Mobile App for Collection of User Experience Data from Mobile Network Subscribers

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1. Introduction

The continuous evolution of mobile networks has increased user expectations regarding service quality, reliability, and performance. In Cameroon, mobile subscribers frequently experience fluctuations in network quality such as slow data speeds, high latency, call drops, and service outages. However, current monitoring approaches used by mobile network operators focus primarily on technical metrics and often fail to capture real user experience (Quality of Experience: QoE).

In this requirement gathering phase, we aimed to bridge this gap by collecting both **subjective user feedback** and **objective network performance data**. We combined survey-based insights with measurements obtained from tools such as the Speedtest application. This report presents the stakeholders involved, techniques used, data collection and cleaning processes, user behavior analysis, and key observations that will guide the design and implementation of the proposed mobile application.

2. Stakeholder Identification

Identifying stakeholders was a crucial first step in our requirement gathering process, it ensured that all relevant perspectives were considered. The stakeholders for this system include:

Primary Stakeholders

- **Mobile Subscribers (Users):** Provide feedback and allow passive data collection
- **Mobile Network Operators (MTN, Orange, Camtel):** Use collected data to improve network performance
- **System Developers (Us):** Design and implement the application

Secondary Stakeholders

- **Regulatory Bodies (e.g., Telecommunications Authorities):** Monitor service quality standards
- **Researchers/Data Analysts:** Use data for QoE analysis and insights

We also considered the importance of **meeting with network providers**, as they are key beneficiaries of the system. These meetings would help align our data collection with industry needs.

3. Requirement Gathering Techniques

We used multiple techniques to ensure comprehensive data collection:

a. Surveys (Primary Method Used)

https://docs.google.com/forms/d/e/1FAIpQLSdyTjhxLZghjgc0bwADUty7n-SeULA_7nCkLEAgnk-r2LNwmQ/viewform?usp=preview

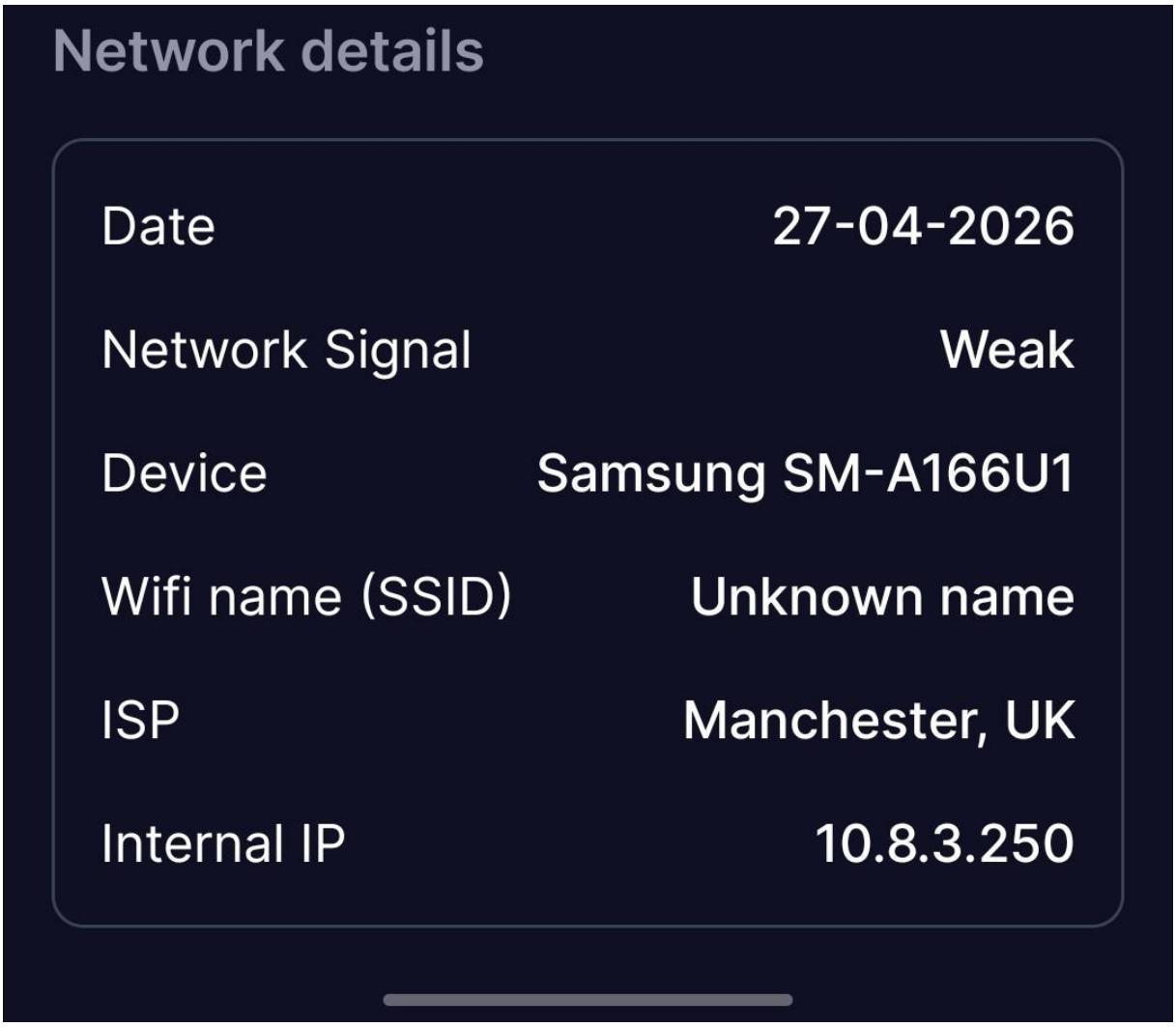
We designed and distributed questionnaires to mobile users to understand and to collect user perceptions regarding:

- Their satisfaction levels
- Common network issues (slow internet, dropped calls)
- Frequency of poor service experiences
- Network reliability
- Internet speed satisfaction
- Frequency of service disruptions
- Overall user experience

Process:

We ask people to download, run speed test app, and send their result to us for analysis.

This is a sample of what we received from users.



We shared the link to our Google form (containing useful questions) to people in the country and we obtained the following results.

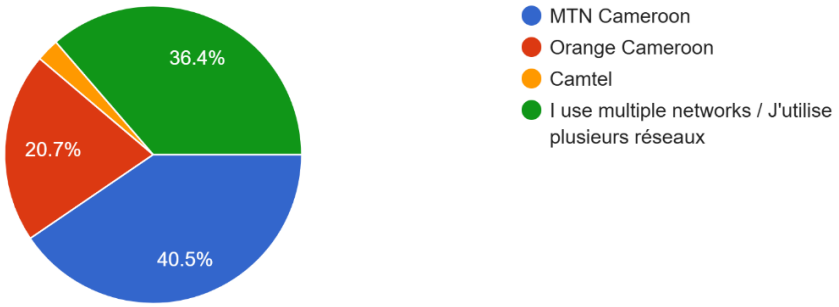
https://docs.google.com/spreadsheets/d/110fy0ZcMNtOjLP353Mb6jg3b5rd3wSPkUIoJRf_X0VA/edit?usp=sharing

2. What is your occupation? /Quelle est votre profession ?

121 responses

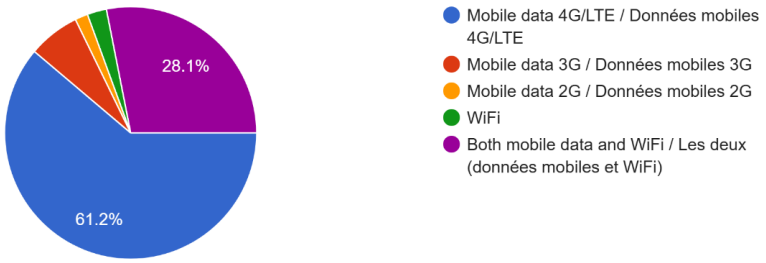
4. Which mobile network operator do you use? / Quel opérateur de réseau mobile utilisez-vous ?

121 responses



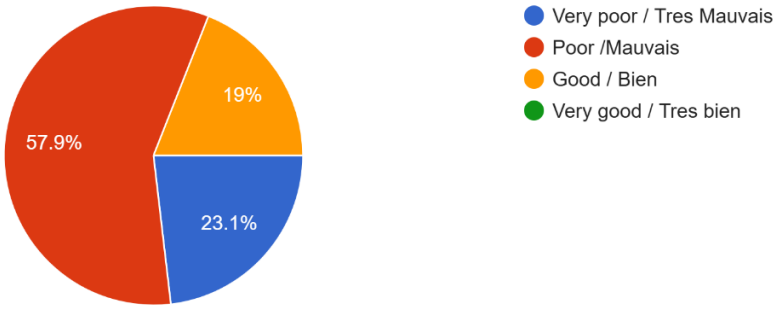
5. What type of internet connection do you mostly use? / Quel type de connexion internet utilisez-vous le plus souvent ?

121 responses



7. How would you rate your overall mobile network experience in Cameroon? / Comment évaluez-vous votre expérience globale du réseau mobile au Cameroun ?

121 responses

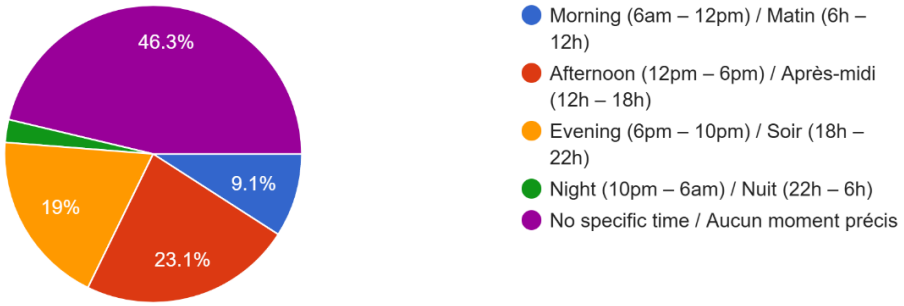


8. How often do you experience poor network quality? / À quelle fréquence rencontrez-vous une mauvaise qualité de réseau ?

121 responses

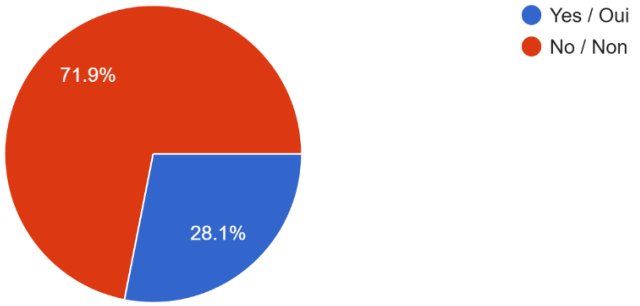
9. At what times do you mostly experience poor network quality? /À quels moments rencontrez-vous le plus souvent une mauvaise qualité de réseau ?

121 responses



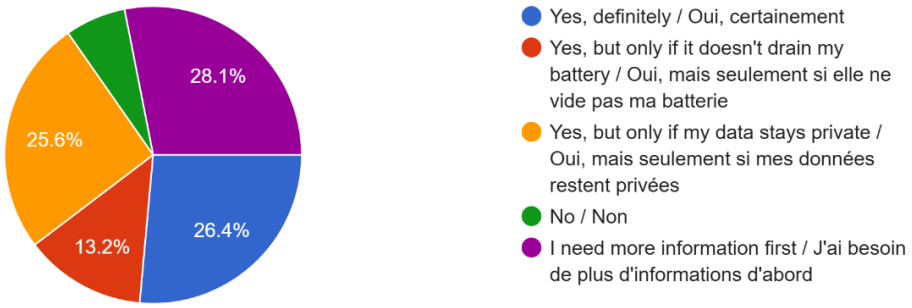
13. Have you ever used a network monitoring app before? / Avez-vous déjà utilisé une application de surveillance réseau auparavant ?

121 responses



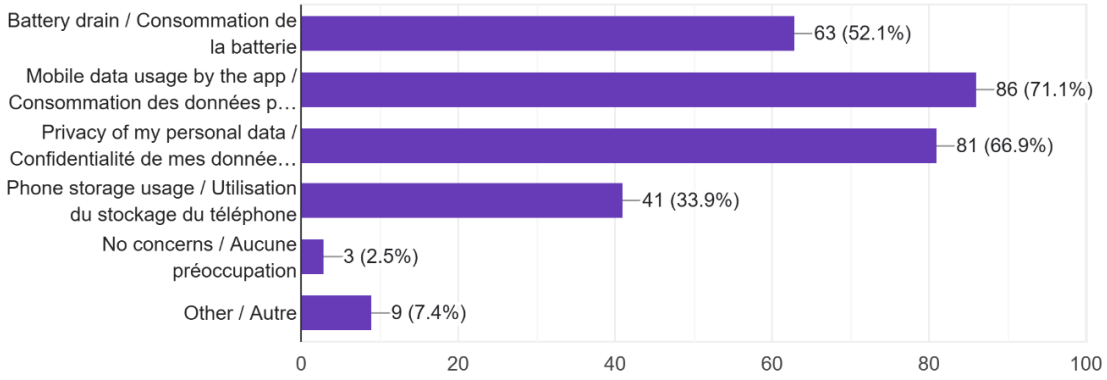
14. Would you be willing to install an app that runs in the background, collects your network quality data, and occasionally asks you to rate your experi...nde occasionnellement d'évaluer votre expérience ?

121 responses



15. What concerns would you have about using such an app? (Select all that apply) / Quelles seraient vos préoccupations concernant l'utilisation... application ? (Sélectionnez tout ce qui s'applique)

121 responses



16. How often would you be comfortable being prompted to rate your network experience? / À quelle fréquence seriez-vous à l'aise pour être invité(e) à évaluer votre expérience réseau ?
121 responses

b. Interviews

We conducted informal interviews with users to gain deeper insights into their expectations and frustrations.

c. Brainstorming

As a team, we brainstormed features such as background data collection & monitoring, real-time alerts, and User rating system.

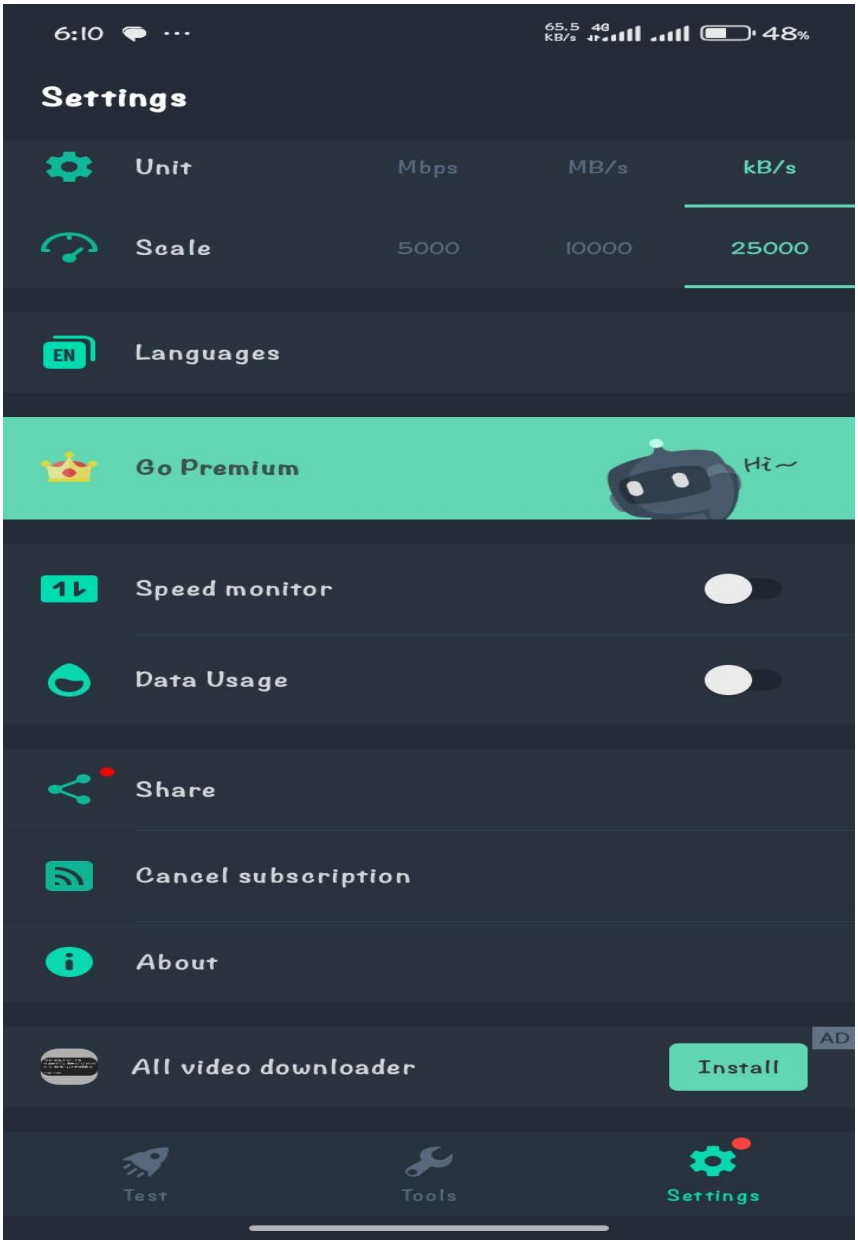
d. Reverse Engineering

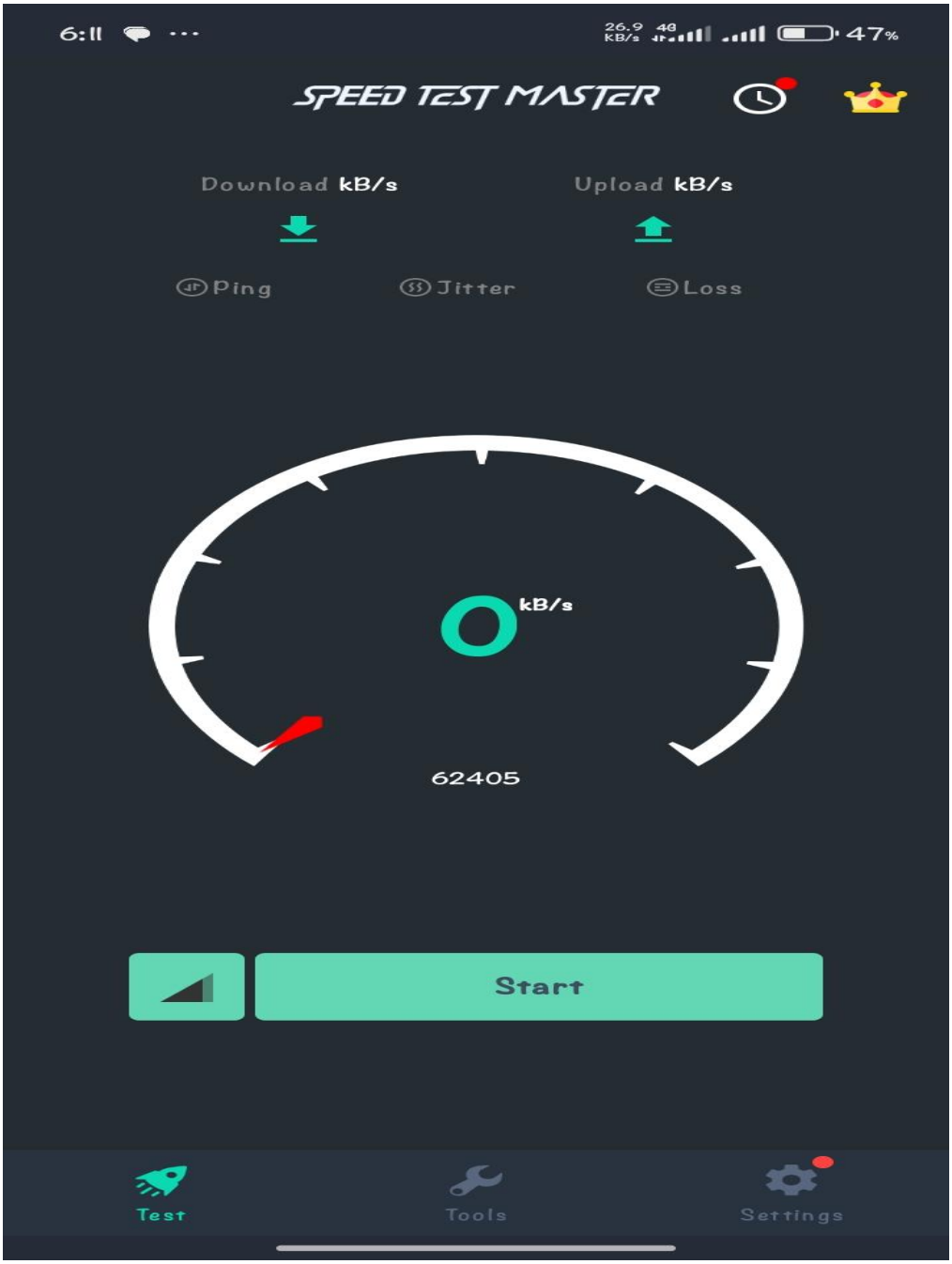
We analyzed existing applications such as:

- **Speedtest by Ookla**
- **OpenSignal**
- **NetMonster**

These platforms helped us understand how network metrics are collected and presented, best practices and limitations.

Here we have some features of the speed test app:


6:10
48%
Settings
Unit
Mbps
MB/s
kB/s
Scale
5000
10000
25000
Languages
Go Premium
Hi~
Speed monitor
Data Usage
Share
Cancel subscription
About
All video downloader
Install
AD
Test
Tools
Settings



4. Data Gathering

Data gathering was carried out using two main approaches:

a. Survey Data

We collected responses from users regarding:

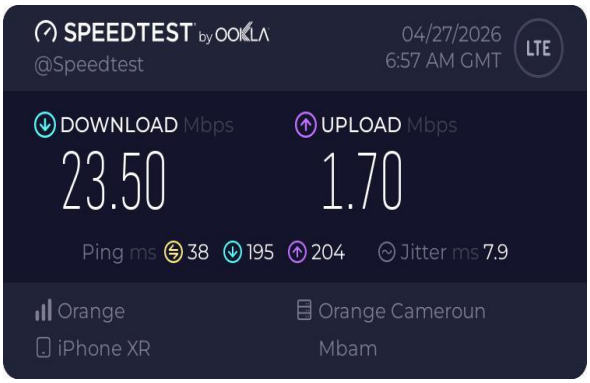
- Network satisfaction
- Common issues experienced (slow speed, call drops)

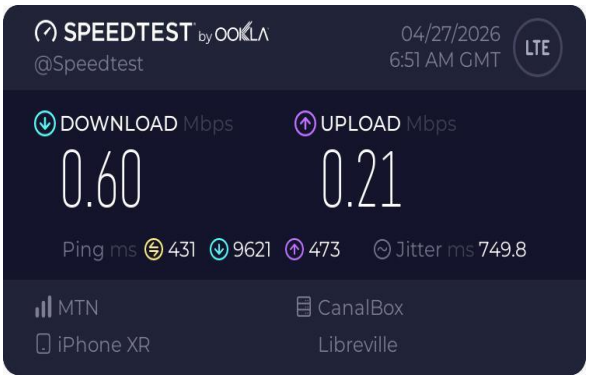
- Preferred features in a mobile app

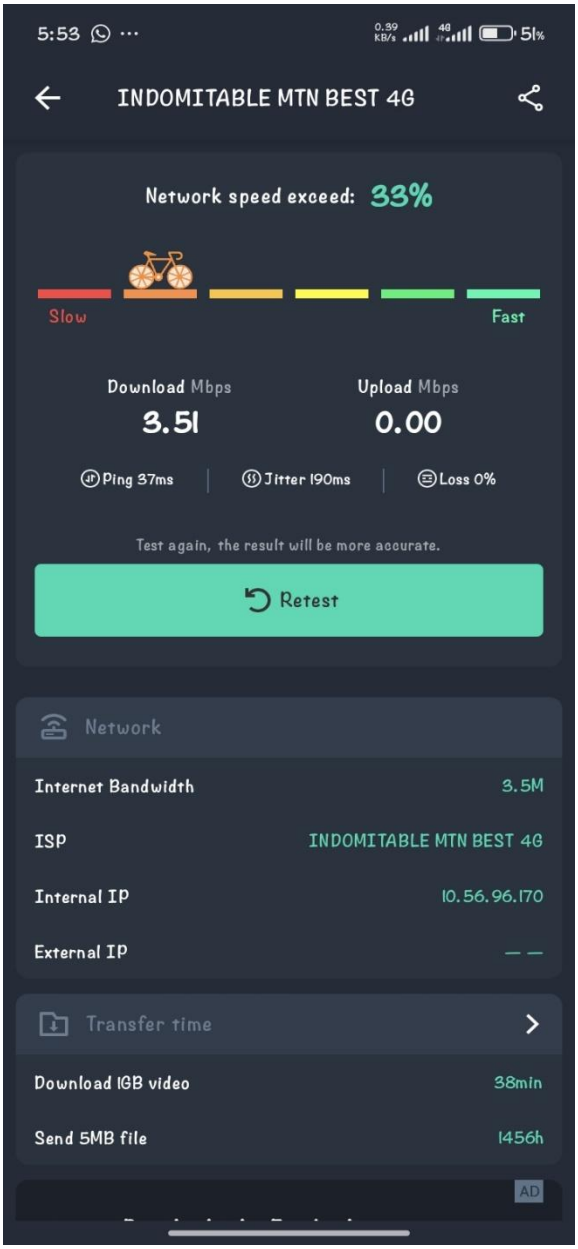
b. Objective Data (Speedtest App Results)

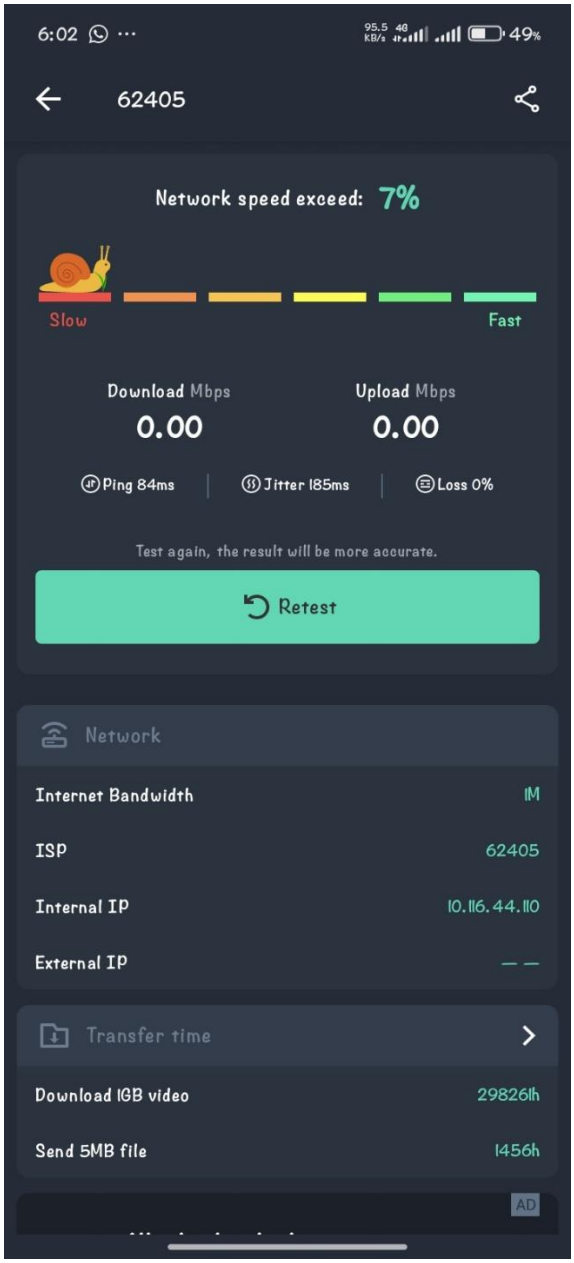
We used results from the **Speedtest application** to collect:

- Download and upload speeds
- Latency (ping)
- Network type (3G, 4G, etc.)









This allowed us to combine **subjective user feedback** with **objective performance metrics**.

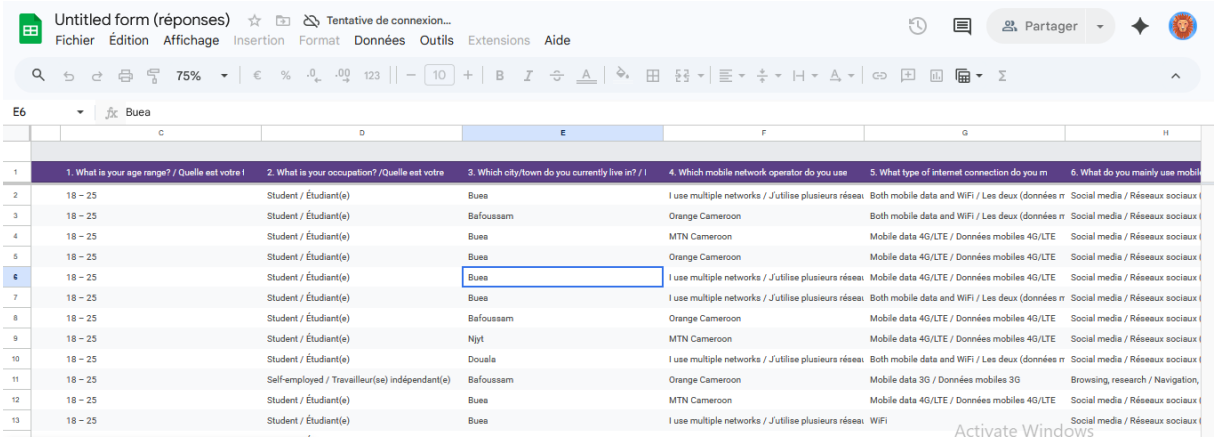
5. Data Cleaning

After collecting the data, we performed data cleaning to ensure accuracy and reliability:

- Removed duplicate survey responses
- Corrected inconsistent entries (e.g., different formats of network types)
- Handled missing values by filtering incomplete responses
- Standardized data formats (dates, numeric values)

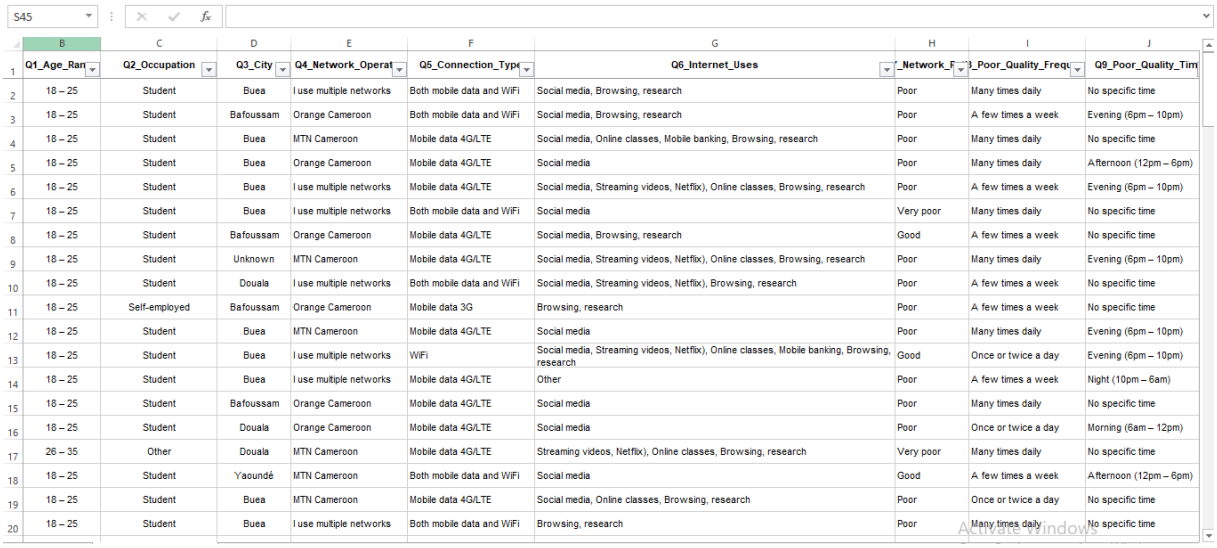
This step is essential to ensure that analysis and system design are based on valid data.

From the duplicated, not standardized, inconsistent and imcomplete responses of the users:



1. What is your age range? / Quelle est votre	2. What is your occupation? /Quelle est votre	3. Which city/town do you currently live in? /	4. Which mobile network operator do you use	5. What type of internet connection do you m	6. What do you mainly use mobil
18 – 25	Student / Étudiant(e)	Buea	I use multiple networks / J'utilise plusieurs réseau	Both mobile data and WiFi / Les deux (données m	Social media / Réseaux sociaux (
18 – 25	Student / Étudiant(e)	Bafoussam	Orange Cameroon	Both mobile data and WiFi / Les deux (données m	Social media / Réseaux sociaux (
18 – 25	Student / Étudiant(e)	Buea	MTN Cameroon	Mobile data 4G/LTE / Données mobiles 4G/LTE	Social media / Réseaux sociaux (
18 – 25	Student / Étudiant(e)	Buea	Orange Cameroon	Mobile data 4G/LTE / Données mobiles 4G/LTE	Social media / Réseaux sociaux (
18 – 25	Student / Étudiant(e)	Buea	I use multiple networks / J'utilise plusieurs réseau	Mobile data 4G/LTE / Données mobiles 4G/LTE	Social media / Réseaux sociaux (
18 – 25	Student / Étudiant(e)	Buea	I use multiple networks / J'utilise plusieurs réseau	Both mobile data and WiFi / Les deux (données m	Social media / Réseaux sociaux (
18 – 25	Student / Étudiant(e)	Bafoussam	Orange Cameroon	Mobile data 4G/LTE / Données mobiles 4G/LTE	Social media / Réseaux sociaux (
18 – 25	Student / Étudiant(e)	Njyt	MTN Cameroon	Mobile data 4G/LTE / Données mobiles 4G/LTE	Social media / Réseaux sociaux (
18 – 25	Student / Étudiant(e)	Douala	I use multiple networks / J'utilise plusieurs réseau	Both mobile data and WiFi / Les deux (données m	Social media / Réseaux sociaux (
18 – 25	Self-employed / Travailleur(se) indépendant(e)	Bafoussam	Orange Cameroon	Mobile data 3G / Données mobiles 3G	Browsing, research / Navigation,
18 – 25	Student / Étudiant(e)	Buea	MTN Cameroon	Mobile data 4G/LTE / Données mobiles 4G/LTE	Social media / Réseaux sociaux (
18 – 25	Student / Étudiant(e)	Buea	I use multiple networks / J'utilise plusieurs réseau	WiFi	Social media / Réseaux sociaux (

To well defined, and cleaned responses:



Q1_Age_Ran	Q2_Occupation	Q3_City	Q4_Network_Operat	Q5_Connection_Type	Q6_Internet_Uses	_Network_F	3_Poor_Quality_Frequ	Q9_Poor_Quality_Tim
18 – 25	Student	Buea	I use multiple networks	Both mobile data and WiFi	Social media, Browsing, research	Poor	Many times daily	No specific time
18 – 25	Student	Bafoussam	Orange Cameroon	Both mobile data and WiFi	Social media, Browsing, research	Poor	A few times a week	Evening (6pm – 10pm)
18 – 25	Student	Buea	MTN Cameroon	Mobile data 4G/LTE	Social media, Online classes, Mobile banking, Browsing, research	Poor	Many times daily	No specific time
18 – 25	Student	Buea	Orange Cameroon	Mobile data 4G/LTE	Social media	Poor	Many times daily	Afternoon (12pm – 6pm)
18 – 25	Student	Buea	I use multiple networks	Mobile data 4G/LTE	Social media, Streaming videos, Netflix), Online classes, Browsing, research	Poor	A few times a week	Evening (6pm – 10pm)
18 – 25	Student	Buea	I use multiple networks	Both mobile data and WiFi	Social media	Very poor	Many times daily	No specific time
18 – 25	Student	Bafoussam	Orange Cameroon	Mobile data 4G/LTE	Social media, Browsing, research	Good	A few times a week	No specific time
18 – 25	Student	Unknown	MTN Cameroon	Mobile data 4G/LTE	Social media, Streaming videos, Netflix), Online classes, Browsing, research	Poor	Many times daily	Evening (6pm – 10pm)
18 – 25	Student	Douala	I use multiple networks	Both mobile data and WiFi	Social media, Streaming videos, Netflix), Browsing, research	Poor	A few times a week	No specific time
18 – 25	Self-employed	Bafoussam	Orange Cameroon	Mobile data 3G	Browsing, research	Poor	A few times a week	No specific time
18 – 25	Student	Buea	MTN Cameroon	Mobile data 4G/LTE	Social media	Poor	Many times daily	Evening (6pm – 10pm)
18 – 25	Student	Buea	I use multiple networks	WiFi	Social media, Streaming videos, Netflix), Online classes, Mobile banking, Browsing, research	Good	Once or twice a day	Evening (6pm – 10pm)
18 – 25	Student	Buea	I use multiple networks	Mobile data 4G/LTE	Other	Poor	A few times a week	Night (10pm – 6am)
18 – 25	Student	Bafoussam	Orange Cameroon	Mobile data 4G/LTE	Social media	Poor	Many times daily	No specific time
18 – 25	Student	Douala	Orange Cameroon	Mobile data 4G/LTE	Social media	Poor	Once or twice a day	Morning (6am – 12pm)
26 – 35	Other	Douala	MTN Cameroon	Mobile data 4G/LTE	Streaming videos, Netflix), Online classes, Browsing, research	Very poor	Many times daily	No specific time
18 – 25	Student	Yaoundé	MTN Cameroon	Both mobile data and WiFi	Social media	Good	A few times a week	Afternoon (12pm – 6pm)
18 – 25	Student	Buea	MTN Cameroon	Mobile data 4G/LTE	Social media, Online classes, Browsing, research	Poor	Once or twice a day	No specific time
18 – 25	Student	Buea	I use multiple networks	Both mobile data and WiFi	Browsing, research	Poor	Many times daily	No specific time

We extracted useful data from the the pictures of the speed test app we received from users and classified them for better analysis and visualization.

B	C	D	E	F	G	H	I
Device type	User ID	Latency(ms)	DL(Mbps)	UL(Mbps)	Bandwidth (Mbps)	Jitter (ms)	Packet Loss
android	690799180	158	16,2	1,86	16,2	121	0
ios	672542768	132	3	3	2.4	198	8
ios	658487098	79	1	1	0.7	79	15
android	620112911	64	9.29	1.79	9.3	112	0
ios	659206984	367	1	1	0.2	--	23
android	697029633	252.1	0.79	3.7	2.5	7.62	0
ios	620173294	143	0.42	0.45	1	48	0
android	699362055	2845	4	2	12.4	_	0
ios	694067501	2536	13	2	12.6		0
android	675080477	140	1.29	0.35	1.3	19	0

6. Observations from Collected Data

From the analysis of survey responses and Speedtest results, we observed that:

- A large percentage of users reported **low satisfaction with network speed**, especially during peak hours.
- Many users experienced **frequent network instability**, including dropped connections and slow browsing.
- There is a noticeable gap between **perceived performance and actual measured speeds**, indicating inconsistency.
- Users expressed strong interest in a **simple app that provides feedback and monitors performance automatically with low battery and data consumption**.
- Speedtest results showed **variability in latency and bandwidth depending on location**, highlighting coverage issues.

These observations confirm the need for a system that continuously collects QoE data.

7. User Reluctance Assessment

In our project to design a mobile application for collecting user experience data from mobile network subscribers in Cameroon, assessing user reluctance was an essential part of the requirement gathering phase. User reluctance refers to the resistance potential users may have toward adopting or continuously using a future application. Since our proposed app will require background operation, location access, and periodic feedback, understanding these reluctance factors now helps us gather the right requirements before any implementation begins. We identified potential resistance from users regarding the use of the application:

Challenges identified:

Reluctance Factor	Description
Battery drain fear	Users may worry that continuous background operation will quickly deplete their phone battery
Data usage concern	Users on limited or expensive data plans may fear unexpected data consumption
Privacy invasion	Collection of location and device metrics could be perceived as surveillance
Annoyance from prompts	Frequent feedback requests could interrupt normal phone activities
Lack of personal benefit	Users may see no reason to help the operator

	without receiving something in return
Low trust in operators	Mobile network operators are not always trusted by subscribers in Cameroon

Solutions:

- Implement clear privacy policies
- Allow users to control data sharing
- Optimize the background processing of the app for low battery usage
- Provide transparency on how data is used
- Clear explanation of benefits

8. Additional Features Proposed

After analysing exixting systems and user responses, to improve the system, we propose the following features:

- Real-time network performance dashboard
- Anonymous data collection option
- Notifications for poor network quality
- Data export for research purposes
- Integration with maps to visualize network coverage (maps, graphs)
- Network issue alerts
- Exportable reports

9. Conclusion

The requirement gathering phase provided valuable insights into user needs, system expectations, identification of key stakeholders, collection of both subjective and objective data necessary for system design, and understading of existing challenges in mobile network performance monitoring. By combining survey responses with objective Speedtest data, we identified key issues such as inconsistent network quality and user dissatisfaction. The findings highlight the importance of developing a mobile application capable of collecting real-time QoE data in an automated and user-friendly manner. Addressing user concerns and integrating advanced features will ensure higher adoption and effectiveness of the system. This phase establishes a strong foundation for the next stages of design and implementation.

10. Reference

https://www.fcc.gov/sites/default/files/2022_fcc_speed_test_app_technical_description.pdf